

1. Logical Representation (1.5 pts)

1. Translate the following English sentences into FOL. Consider predicates

$\text{Person}(x), \text{Likes}(x, y), \text{Older}(x, y), \text{Happy}(x), \text{Friend}(x, y), \text{Teacher}(x, y)$

- (a) Everyone likes John.
 - (b) No one is older than themselves.
 - (c) All friends of John are happy.
 - (d) If someone is a teacher of Tim, then that teacher is older than Tim.
 - (e) There is a person who likes everyone who likes them.
2. Translate each FOL sentence into an English sentence.

- (a) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Friend}(x, y) \wedge \text{Happy}(y)))$
- (b) $\exists x (\text{Person}(x) \wedge \forall y (\text{Person}(y) \rightarrow \text{Older}(x, y)))$
- (c) $\forall x (\text{Person}(x) \rightarrow (\text{Friend}(\text{John}, x) \leftrightarrow \text{Likes}(x, \text{John})))$
- (d) $\exists x (\text{Person}(x) \wedge \text{Likes}(x, \text{Tim}) \wedge \forall y ((\text{Person}(y) \wedge \text{Likes}(y, x)) \rightarrow \text{Happy}(y)))$
- (e) $\forall x (\text{Person}(x) \rightarrow (\exists y (\text{Teacher}(y, x) \wedge \text{Older}(y, x)) \rightarrow \text{Happy}(x)))$

Solution:

1. (a) Everyone likes John: $\forall x (\text{Person}(x) \rightarrow \text{Likes}(x, \text{John}))$
(b) No one is older than themselves: $\forall x (\text{Person}(x) \rightarrow \neg \text{Older}(x, x))$,
or equivalently $\neg \exists x (\text{Person}(x) \wedge \text{Older}(x, x))$.
(c) All friends of John are happy.

$\forall x ((\text{Person}(x) \wedge \text{Friend}(x, \text{John})) \rightarrow \text{Happy}(x))$

- (d) If someone is a teacher of Tim, then that teacher is older than Tim.

$\forall x ((\text{Person}(x) \wedge \text{Teacher}(x, \text{Tim})) \rightarrow \text{Older}(x, \text{Tim}))$

- (e) There is a person who likes everyone who likes them.

$\exists x (\text{Person}(x) \wedge \forall y (\text{Person}(y) \wedge \text{Likes}(y, x) \rightarrow \text{Likes}(x, y)))$

2. (a) $\forall x (\text{Person}(x) \rightarrow \exists y (\text{Friend}(x, y) \wedge \text{Happy}(y)))$
Every person has a friend who is happy.
(b) $\exists x (\text{Person}(x) \wedge \forall y (\text{Person}(y) \rightarrow \text{Older}(x, y)))$
There is a person who is older than every person.
(c) $\forall x (\text{Person}(x) \rightarrow (\text{Friend}(\text{John}, x) \leftrightarrow \text{Likes}(x, \text{John})))$
For every person, that person is a friend of John if and only if that person likes John.
(d) $\exists x (\text{Person}(x) \wedge \text{Likes}(x, \text{Tim}) \wedge \forall y ((\text{Person}(y) \wedge \text{Likes}(y, x)) \rightarrow \text{Happy}(y)))$
There exists a person who likes Tim, and every person who likes that person is happy.
(e) $\forall x (\text{Person}(x) \rightarrow (\exists y (\text{Teacher}(y, x) \wedge \text{Older}(y, x)) \rightarrow \text{Happy}(x)))$
Every person who has a teacher that is older than them is happy.

2. Logical Inference (1.5 pts)

- Given the following definite clause knowledge base KB, apply SLD resolution to prove that

$$KB \models \text{can_rent}(\text{Company2}, \text{John}, \text{Car1})$$

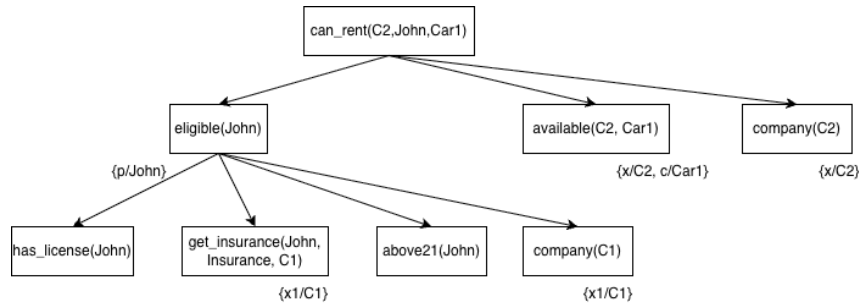
Provide the substitutions resulting from unification at each step. Draw the search tree of the SLD resolution process.

Clause
1 has_license(John)
2 get_insurance(John, Insurance, Company1)
3 above21(John)
4 available(Company2, Car1)
5 company(Company1)
6 company(Company2)
7 $\forall p, x : \text{eligible}(p) \leftarrow \text{has_license}(p) \wedge \text{get_insurance}(p, \text{Insurance}, x) \wedge \text{above21}(p) \wedge \text{company}(x)$
8 $\forall x, p, c : \text{can_rent}(x, p, c) \leftarrow \text{eligible}(p) \wedge \text{available}(x, c) \wedge \text{company}(x)$

Solution:

Step	Goal to resolve	Unification	Remaining Goals
1	$\text{can_rent}(x, p, c)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}\}$	$\text{eligible}(p), \text{available}(c), \text{company}(x)$
2	$\text{eligible}(p)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}\}$	$\text{available}(x, c), \text{company}(x), \text{has_license}(p), \text{get_insurance}(p, \text{Insurance}, x_1), \text{above21}(p), \text{company}(x_1)$
3	$\text{available}(x, c)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}\}$	$\text{company}(x), \text{has_license}(p), \text{get_insurance}(p, \text{Insurance}, x_1), \text{above21}(p), \text{company}(x_1)$
4	$\text{company}(x)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}\}$	$\text{has_license}(p), \text{get_insurance}(p, \text{Insurance}, x_1), \text{above21}(p), \text{company}(x_1)$
5	$\text{has_license}(p)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}\}$	$\text{get_insurance}(p, \text{Insurance}, x_1), \text{above21}(p), \text{company}(x_1)$
6	$\text{get_insurance}(p, \text{Insurance}, x_1)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}, x_1/\text{Company1}\}$	$\text{above21}(p), \text{company}(x_1)$
7	$\text{above21}(p)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}, x_1/\text{Company1}\}$	$\text{company}(x_1)$
8	$\text{company}(x_1)$	$\{x/\text{Company2}, p/\text{John}, c/\text{Car1}, x_1/\text{Company1}\}$	

The process resolves all goals and thus we can proof $KB \models \text{can_rent}(\text{Company2}, \text{John}, \text{Car1})$, with unification: $\{x/\text{Company2}, p/\text{John}, c/\text{Car1}, x_1/\text{Company1}\}$. The search tree is:



3. Bayesian Networks (2 pts)

Consider a Bayesian network containing the following binary variables (Y or N) and CPT:

- G : Game genre is Action
- E : Player is experienced
- D : Game is difficult
- T : Playtime > 10 hours/week
- F : High enjoyment

CPTs:

$$P(G = Y) = 0.6, \quad P(E = Y) = 0.4$$

G	E	D	$P(D \mid G, E)$		D	T	$P(T \mid D)$
N	N	N	0.8		N	N	0.7
N	N	Y	0.2		N	Y	0.3
N	Y	N	0.5		Y	N	0.2
N	Y	Y	0.5		Y	Y	0.8
Y	N	N	0.3				
Y	N	Y	0.7				
Y	Y	N	0.1				
Y	Y	Y	0.9				

T	E	F	$P(F \mid T, E)$
N	N	N	0.9
N	N	Y	0.1
N	Y	N	0.6
N	Y	Y	0.4
Y	N	N	0.4
Y	N	Y	0.6
Y	Y	N	0.1
Y	Y	Y	0.9

Derive the following probabilities:

- Compute $P(E = Y \mid F = N)$.
- Compute $P(D = Y \mid T = Y, E = Y)$.

Solution:

1. **Restriction of $F=N$:**

T	E	$g_{F=N}(T, E)$
N	N	0.9
N	Y	0.6
Y	N	0.4
Y	Y	0.1

2. **Eliminate G, D, T one by one:**

i. Eliminate G by multiplying $P(G)$ and $P(D|G, E)$:

G	E	D	$P(D G, E)$
N	N	N	$0.8 \times 0.4 = 0.32$
N	N	Y	$0.2 \times 0.4 = 0.08$
N	Y	N	$0.5 \times 0.4 = 0.2$
N	Y	Y	$0.5 \times 0.4 = 0.2$
Y	N	N	$0.3 \times 0.6 = 0.18$
Y	N	Y	$0.7 \times 0.6 = 0.42$
Y	Y	N	$0.1 \times 0.6 = 0.06$
Y	Y	Y	$0.9 \times 0.6 = 0.54$

Sum over G and get $P(D|E)$:

E	D	$P(D E)$
N	N	$0.32 + 0.18 = 0.5$
N	Y	$0.08 + 0.42 = 0.5$
Y	N	$0.2 + 0.06 = 0.26$
Y	Y	$0.2 + 0.54 = 0.74$

ii. Eliminate D by multiplying $P(D|E)$ and $P(T|D)$:

E	D	T	$P(D E)P(T D)$
N	N	N	$0.5 \times 0.7 = 0.35$
N	N	Y	$0.5 \times 0.3 = 0.15$
N	Y	N	$0.5 \times 0.2 = 0.1$
N	Y	Y	$0.5 \times 0.8 = 0.4$
Y	N	N	$0.26 \times 0.7 = 0.182$
Y	N	Y	$0.26 \times 0.3 = 0.078$
Y	Y	N	$0.74 \times 0.2 = 0.148$
Y	Y	Y	$0.74 \times 0.8 = 0.592$

Sum over D:

E	T	$P(T E)$
N	N	$0.35 + 0.1 = 0.45$
N	Y	$0.15 + 0.4 = 0.55$
Y	N	$0.182 + 0.148 = 0.33$
Y	Y	$0.078 + 0.592 = 0.67$

iii. Eliminate T by multiplying $P(T|E)$ and $g_{F=N}(T, E)$:

E	T	$P(T E)g_{F=N}(T, E)$
N	N	$0.45 \times 0.9 = 0.405$
N	Y	$0.55 \times 0.4 = 0.22$
Y	N	$0.33 \times 0.6 = 0.198$
Y	Y	$0.67 \times 0.1 = 0.067$

Sum over T:

E	$h(E)$
N	$0.405 + 0.22 = 0.625$
Y	$0.198 + 0.067 = 0.265$

3. Multiple $h(E)$ with $P(E)$ and normalize the result:

$$P(E = Y | F = N) = \frac{0.265 \times 0.4}{0.625 \times 0.6 + 0.265 \times 0.4} \approx 22.04\%$$

(b) Note that D is independent of F given T and E, we do not need to use the table $P(F|T, E)$.

1. Restriction of $T = Y, E = Y$:

G	D	$f_{E=Y}(D, G)$	D	$g_{T=Y}(D)$
N	N	0.5	N	0.3
N	Y	0.5	Y	0.8
Y	N	0.1		
Y	Y	0.9		

2. Only need to eliminate G:

G	D	$f_{E=Y}(D, G)$		G	$P(G)$		G	D	$f_{E=Y}(D, G)P(G)$
N	N	0.5		N	0.4		N	N	$0.5 \times 0.4 = 0.2$
N	Y	0.5	$\times$	N	0.4	$=$	N	Y	$0.5 \times 0.4 = 0.2$
Y	N	0.1		Y	0.6		Y	N	$0.1 \times 0.6 = 0.06$
Y	Y	0.9					Y	Y	$0.9 \times 0.6 = 0.54$

Sum over G and get:

D	$h(D)$
N	0.26
Y	0.74

Multiply $h(D)$ with $g_{T=Y}(D)$:

D	$h(D)$		D	$g_{T=Y}(D)$		D	$h(D)g_{T=Y}(D)$
N	0.26	$\times$	N	0.3	$=$	N	$0.26 \times 0.3 = 0.078$
Y	0.74		Y	0.8		Y	$0.74 \times 0.8 = 0.592$

Normalize and get $P(D = Y|T = Y, E = Y) = \frac{0.592}{0.592+0.078} \approx 88.36\%$.